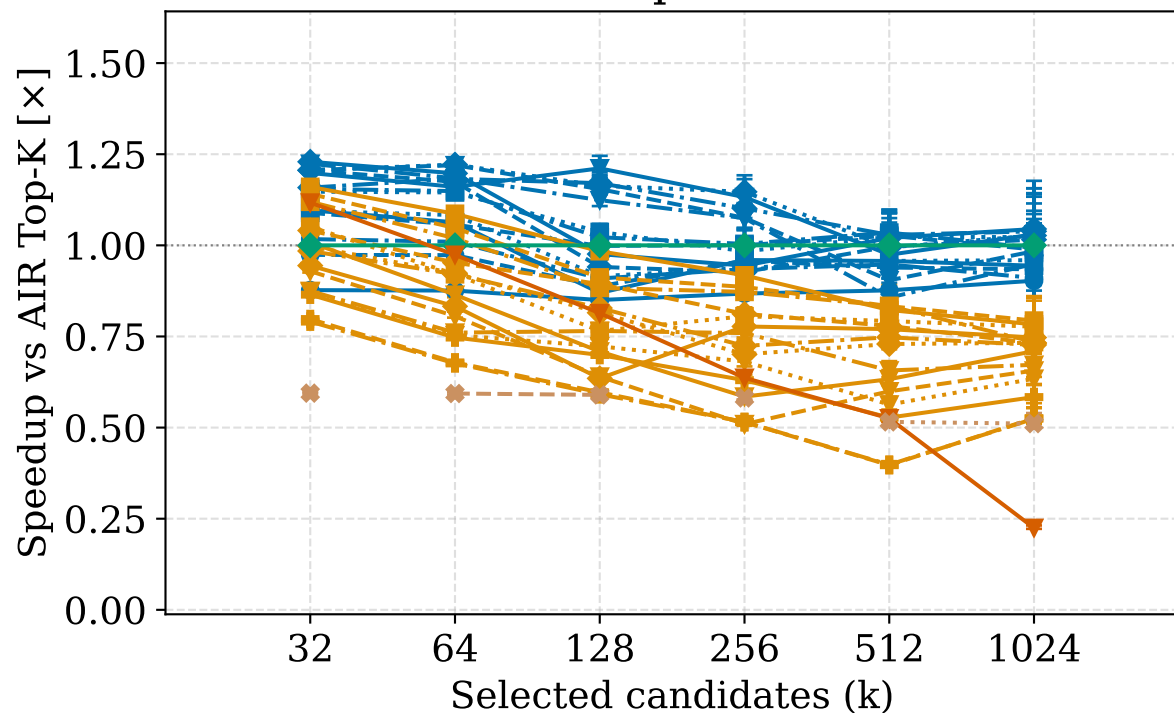
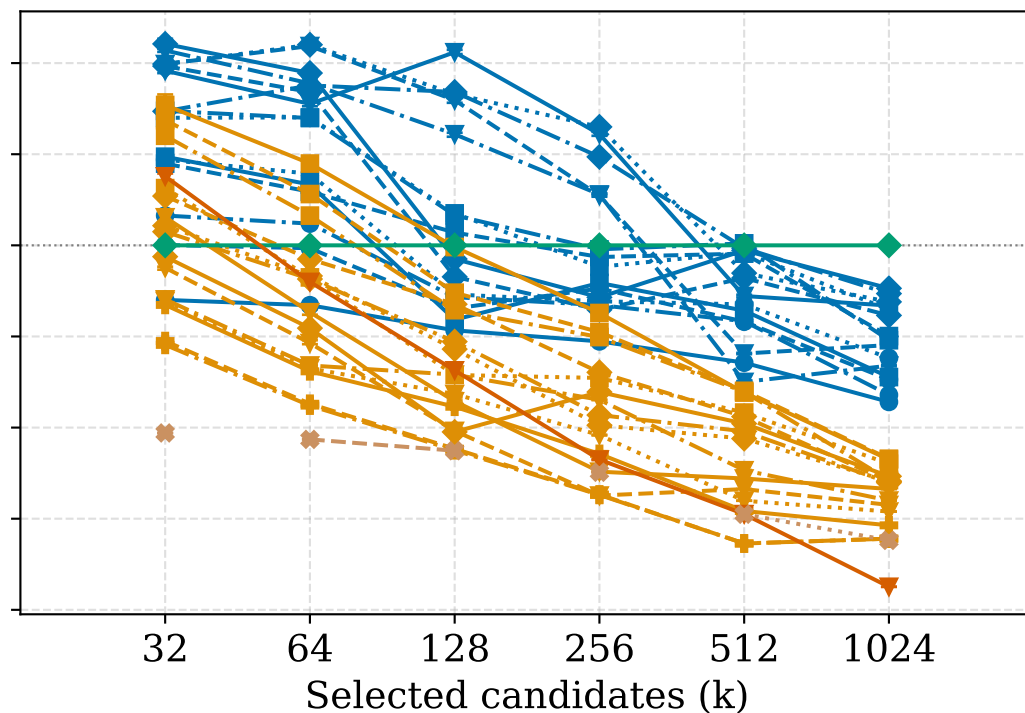


token-sampling: 50,257 candidates; batch=128; input 38c661ffe4; split degree=8; temperature=1.0; seed=42
token-sampling-middle — AIR median / backend median; >1 is faster
Bars: backend IQR; AIR median fixed

Full operator



Isolated selection



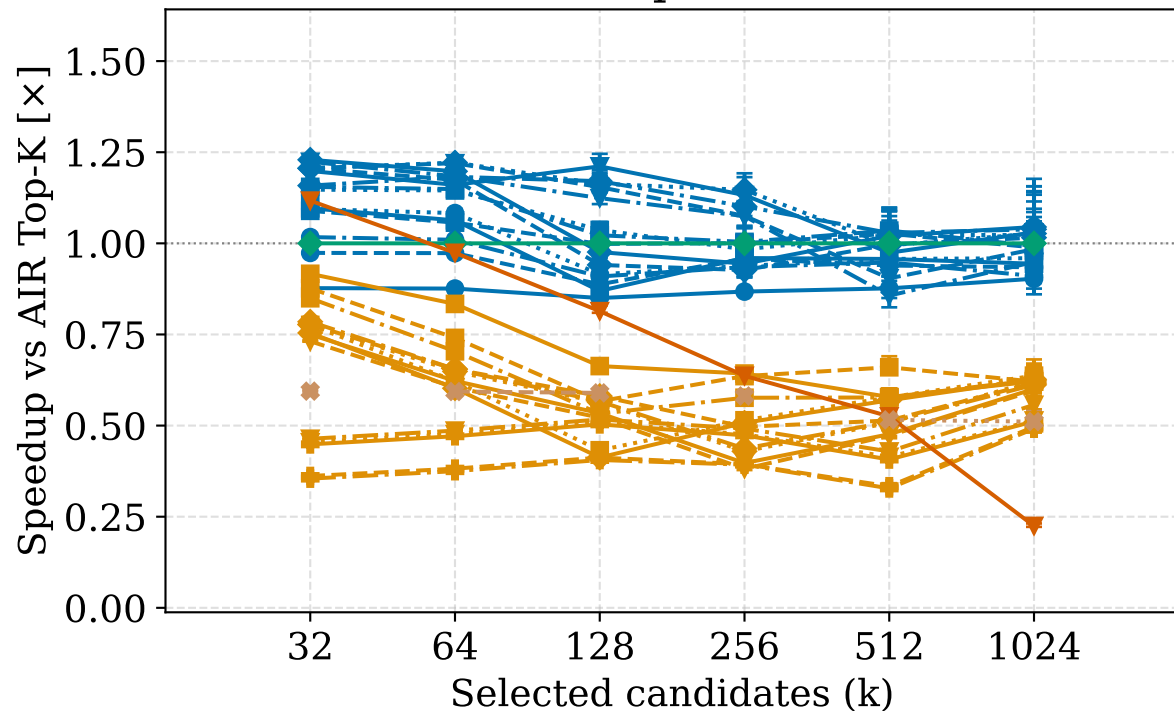
- bits (degree=1, block=128, items=4)
- bits (degree=1, block=128, items=7)
- bits (degree=1, block=128, items=8)
- bits (degree=1, block=128, items=13)
- bits (degree=1, block=128, items=16)
- bits (degree=1, block=256, items=4)
- bits (degree=1, block=256, items=7)
- bits (degree=1, block=256, items=8)
- bits (degree=1, block=256, items=13)
- bits (degree=1, block=256, items=16)
- bits (degree=1, block=512, items=4)
- bits (degree=1, block=512, items=7)
- bits (degree=1, block=512, items=8)
- bits (degree=1, block=512, items=13)
- bits (degree=1, block=512, items=16)
- bits (split) (degree=8, block=128, items=4)
- bits (split) (degree=8, block=128, items=7)
- bits (split) (degree=8, block=128, items=8)
- bits (split) (degree=8, block=256, items=4)
- bits (split) (degree=8, block=256, items=7)
- bits (split) (degree=8, block=256, items=8)
- bits (split) (degree=8, block=256, items=13)
- bits (split) (degree=8, block=256, items=16)
- bits (split) (degree=8, block=512, items=4)
- bits (split) (degree=8, block=512, items=7)
- bits (split) (degree=8, block=512, items=8)
- bits (split) (degree=8, block=512, items=13)
- bits (split) (degree=8, block=512, items=16)
- AIR Top-K (degree=1, block=512, items=0)
- GridSelect (degree=1, block=0, items=0)
- BlockSelect (degree=1, block=128, items=2)
- BlockSelect (degree=1, block=128, items=3)
- BlockSelect (degree=1, block=128, items=4)
- BlockSelect (degree=1, block=128, items=8)

token-sampling: 50,257 candidates; batch=128; input 38c661ffe4; split degree=32; temperature=1.0; seed=42

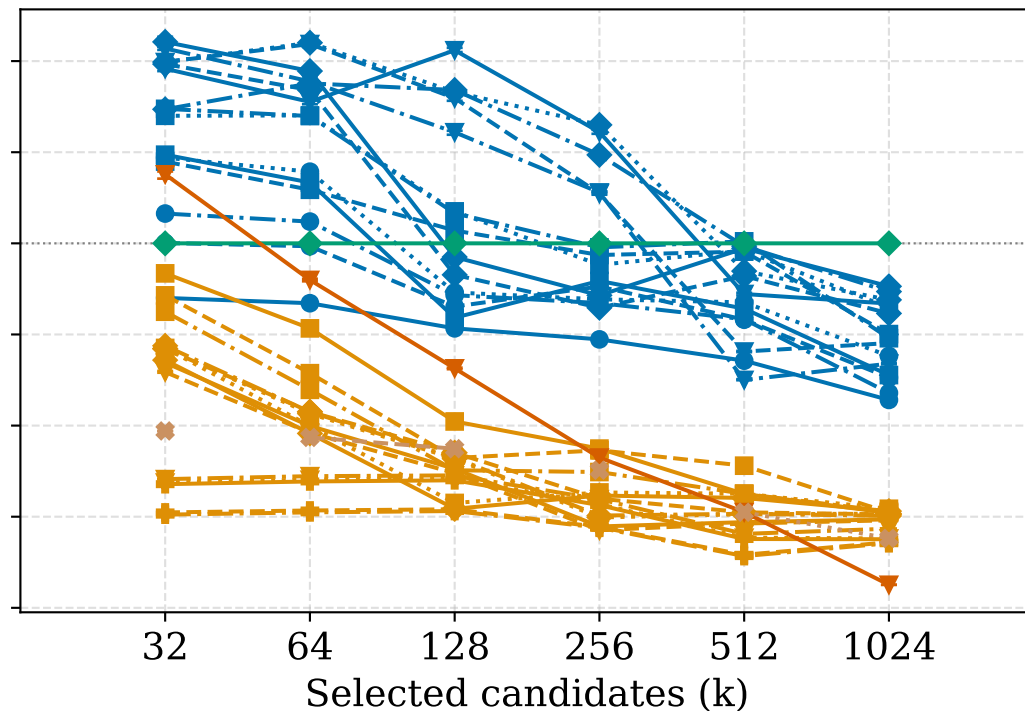
token-sampling-middle — AIR median / backend median; >1 is faster

Bars: backend IQR; AIR median fixed

Full operator



Isolated selection



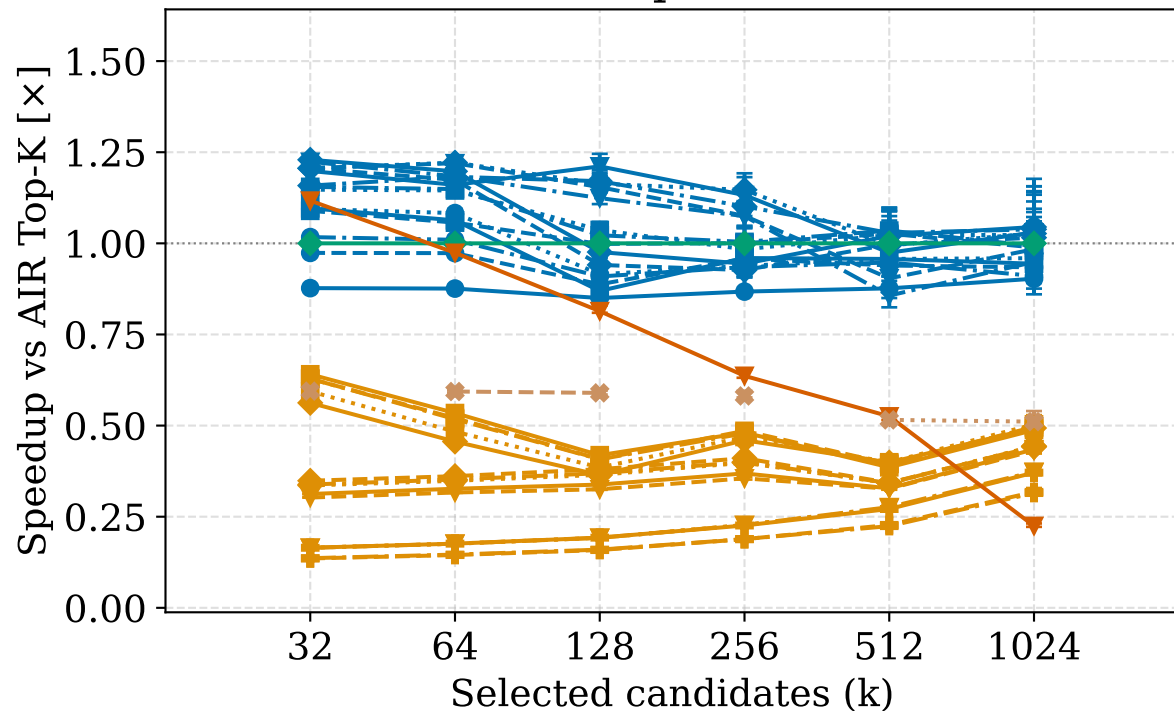
- | | |
|--|---|
| —●— bits (degree=1, block=128, items=4) | —■— bits (split) (degree=32, block=128, items=13) |
| —●— bits (degree=1, block=128, items=7) | —◆— bits (split) (degree=32, block=128, items=16) |
| —●— bits (degree=1, block=128, items=8) | —◆— bits (split) (degree=32, block=256, items=4) |
| —●— bits (degree=1, block=128, items=13) | —◆— bits (split) (degree=32, block=256, items=7) |
| —■— bits (degree=1, block=128, items=16) | —◆— bits (split) (degree=32, block=256, items=8) |
| —■— bits (degree=1, block=256, items=4) | —▼— bits (split) (degree=32, block=256, items=13) |
| —■— bits (degree=1, block=256, items=7) | —▼— bits (split) (degree=32, block=256, items=16) |
| —■— bits (degree=1, block=256, items=8) | —▼— bits (split) (degree=32, block=512, items=4) |
| —◆— bits (degree=1, block=256, items=13) | —▼— bits (split) (degree=32, block=512, items=7) |
| —◆— bits (degree=1, block=256, items=16) | —▼— bits (split) (degree=32, block=512, items=8) |
| —◆— bits (degree=1, block=512, items=4) | —▼— bits (split) (degree=32, block=512, items=13) |
| —◆— bits (degree=1, block=512, items=7) | —▼— bits (split) (degree=32, block=512, items=16) |
| —▼— bits (degree=1, block=512, items=8) | —◆— AIR Top-K (degree=1, block=512, items=0) |
| —▼— bits (degree=1, block=512, items=13) | —▼— GridSelect (degree=1, block=0, items=0) |
| —▼— bits (degree=1, block=512, items=16) | —●— BlockSelect (degree=1, block=128, items=2) |
| —■— bits (split) (degree=32, block=128, items=4) | —●— BlockSelect (degree=1, block=128, items=3) |
| —■— bits (split) (degree=32, block=128, items=7) | —●— BlockSelect (degree=1, block=128, items=4) |
| —■— bits (split) (degree=32, block=128, items=8) | —●— BlockSelect (degree=1, block=128, items=8) |

token-sampling: 50,257 candidates; batch=128; input 38c661ffe4; split degree=128; temperature=1.0; seed=42

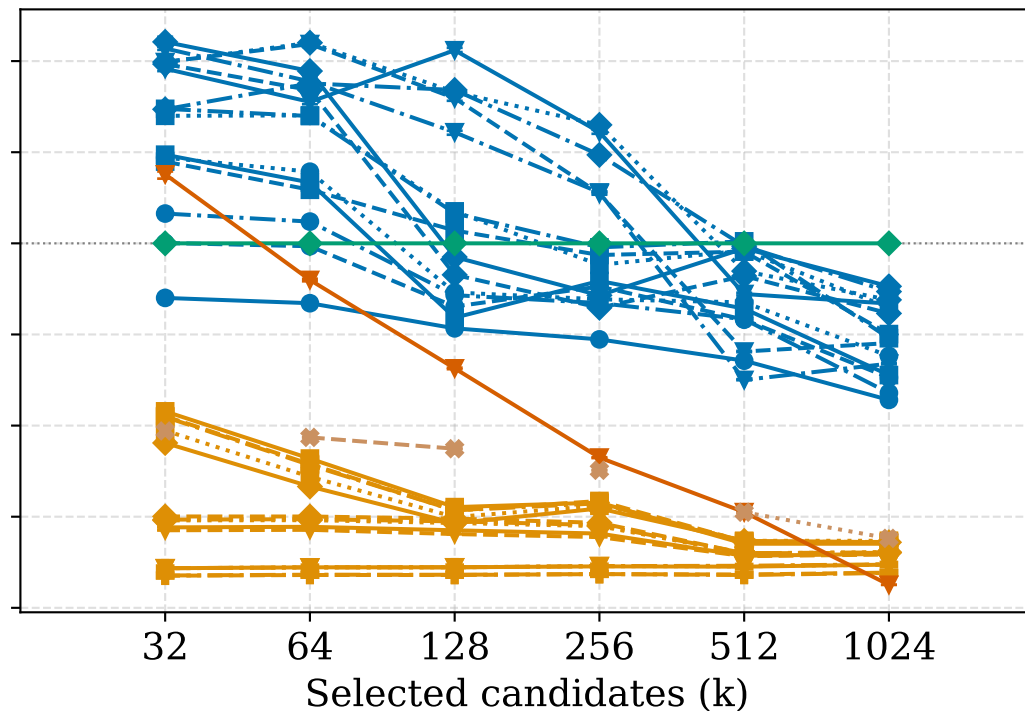
token-sampling-middle — AIR median / backend median; >1 is faster

Bars: backend IQR; AIR median fixed

Full operator



Isolated selection



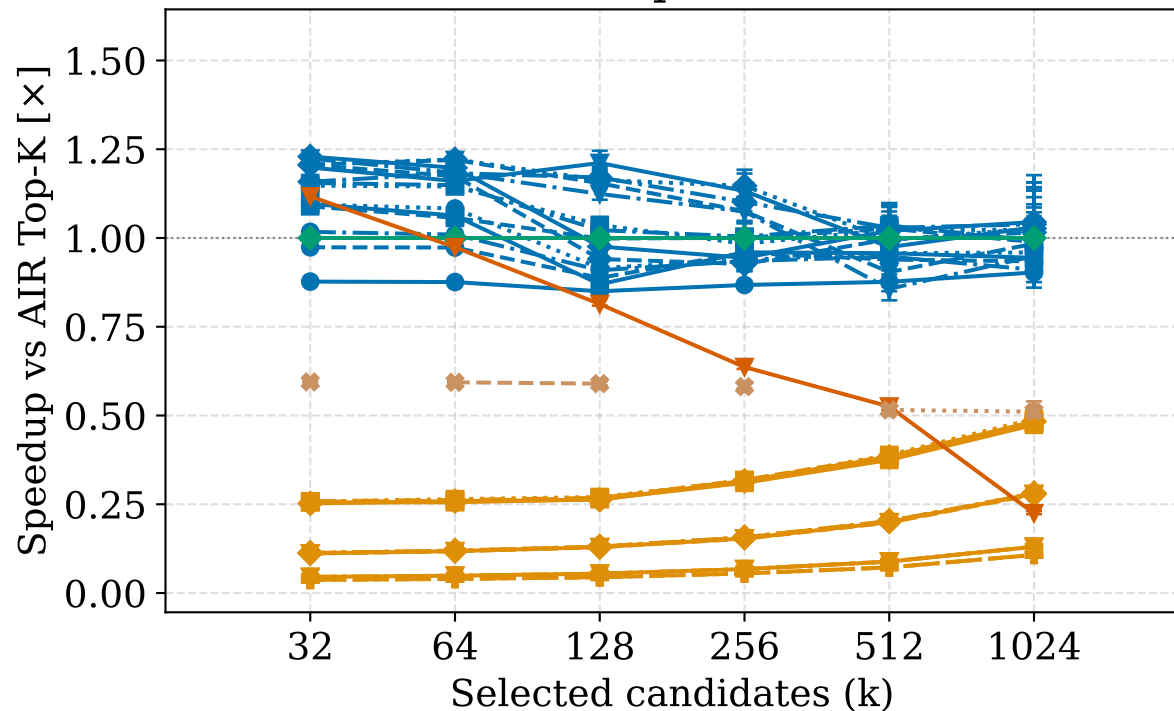
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- bits (degree=1, block=128, items=7)
- bits (degree=1, block=128, items=8)
- bits (degree=1, block=128, items=13)
- bits (degree=1, block=128, items=16)
- bits (degree=1, block=256, items=4)
- bits (degree=1, block=256, items=7)
- bits (degree=1, block=256, items=8)
- bits (degree=1, block=256, items=13)
- bits (degree=1, block=256, items=16)
- bits (degree=1, block=512, items=4)
- bits (degree=1, block=512, items=7)
- bits (degree=1, block=512, items=8)
- bits (degree=1, block=512, items=13)
- bits (degree=1, block=512, items=16)
- bits (split) (degree=128, block=128, items=4)
- bits (split) (degree=128, block=128, items=7)
- bits (split) (degree=128, block=128, items=8)
- bits (split) (degree=128, block=256, items=4)
- bits (split) (degree=128, block=256, items=7)
- bits (split) (degree=128, block=256, items=8)
- bits (split) (degree=128, block=256, items=13)
- bits (split) (degree=128, block=256, items=16)
- bits (split) (degree=128, block=512, items=4)
- bits (split) (degree=128, block=512, items=7)
- bits (split) (degree=128, block=512, items=8)
- bits (split) (degree=128, block=512, items=13)
- bits (split) (degree=128, block=512, items=16)
- AIR Top-K (degree=1, block=512, items=0)
- GridSelect (degree=1, block=0, items=0)
- BlockSelect (degree=1, block=128, items=2)
- BlockSelect (degree=1, block=128, items=3)
- BlockSelect (degree=1, block=128, items=4)
- BlockSelect (degree=1, block=128, items=8)

token-sampling: 50,257 candidates; batch=128; input 38c661ffe4; split degree=512; temperature=1.0; seed=42

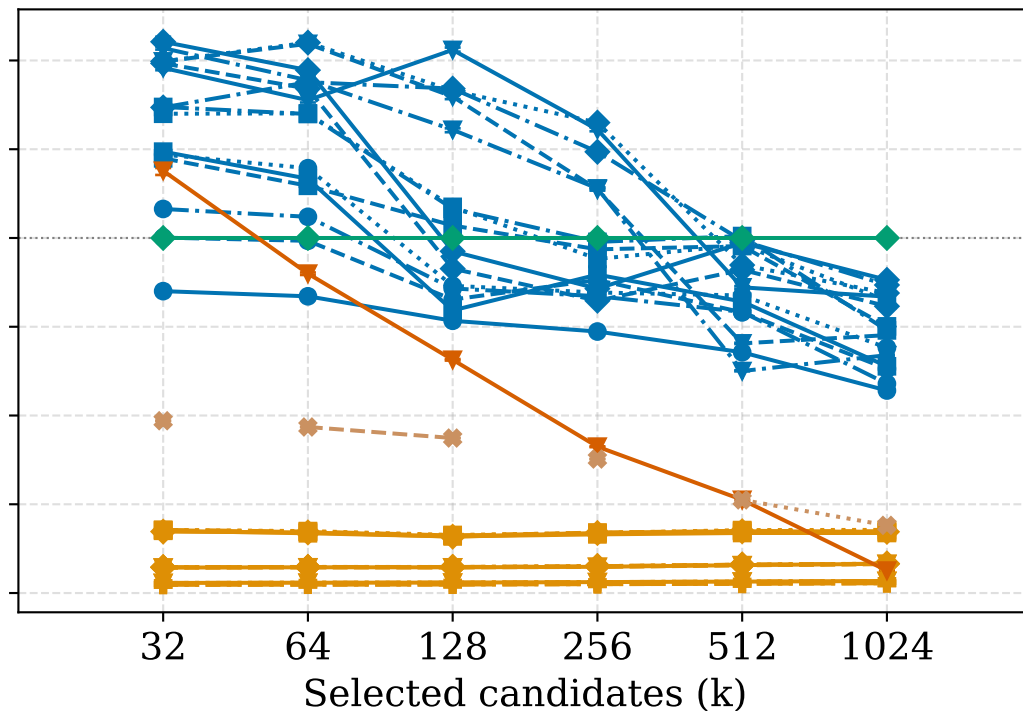
token-sampling-middle — AIR median / backend median; >1 is faster

Bars: backend IQR; AIR median fixed

Full operator



Isolated selection



- bits (degree=1, block=128, items=4)
- bits (degree=1, block=128, items=7)
- bits (degree=1, block=128, items=8)
- bits (degree=1, block=128, items=13)
- bits (degree=1, block=128, items=16)
- bits (degree=1, block=256, items=4)
- bits (degree=1, block=256, items=7)
- bits (degree=1, block=256, items=8)
- bits (degree=1, block=256, items=13)
- bits (degree=1, block=256, items=16)
- bits (degree=1, block=512, items=4)
- bits (degree=1, block=512, items=7)
- bits (degree=1, block=512, items=8)
- bits (degree=1, block=512, items=13)
- bits (degree=1, block=512, items=16)
- bits (split) (degree=512, block=128, items=4)
- bits (split) (degree=512, block=128, items=7)
- bits (split) (degree=512, block=128, items=8)
- bits (split) (degree=512, block=256, items=4)
- bits (split) (degree=512, block=256, items=7)
- bits (split) (degree=512, block=256, items=8)
- bits (split) (degree=512, block=256, items=13)
- bits (split) (degree=512, block=256, items=16)
- bits (split) (degree=512, block=512, items=4)
- bits (split) (degree=512, block=512, items=7)
- bits (split) (degree=512, block=512, items=8)
- bits (split) (degree=512, block=512, items=13)
- bits (split) (degree=512, block=512, items=16)
- AIR Top-K (degree=1, block=512, items=0)
- GridSelect (degree=1, block=0, items=0)
- BlockSelect (degree=1, block=128, items=2)
- BlockSelect (degree=1, block=128, items=3)
- BlockSelect (degree=1, block=128, items=4)
- BlockSelect (degree=1, block=128, items=8)